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Harnessing rainwater for dryland agriculture: a review of harvesting and managed recharge strategies

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Arid and semi-arid regions, encompassing nearly half of Earth's land surface, face escalating water scarcity due to precipitation variability, high evaporative demand, and climate change-induced extremes. While groundwater has historically buffered surface water deficits, widespread overexploitation has precipitated aquifer depletion, land subsidence, and water quality deterioration. Conventional irrigation practices—characterized by low water-use efficiency and weak governance—exacerbate these pressures. This review synthesizes principles and multi-scale applications of rainwater harvesting (RWH) as a strategic complement to conventional water management. It examines technical synergies between decentralized RWH systems and managed aquifer recharge (MAR), demonstrating their combined potential to alleviate stress on overexploited groundwater reserves. We address four objectives: (1) analyzing how increasing precipitation variability and extreme climatic events destabilize agricultural systems and exacerbate water risk; (2) evaluating technical and managerial inefficiencies in conventional irrigation; (3) highlighting cascading environmental, economic, and social costs of groundwater overdraft; and (4) proposing an integrated RWH–MAR framework to enhance water availability, improve ecosystem resilience, and contribute to sustainable agricultural intensification. Our findings show that integrating RWH with efficient irrigation, agroecological practices, and supportive governance can significantly improve water-use efficiency and system resilience. Advancing this approach requires a paradigm shift treating rainwater as a managed resource within a unified hydrological cycle. We conclude that revitalizing rainwater harvesting constitutes a cornerstone for building sustainable agricultural systems and ensuring food security under increasing climatic uncertainty. Key advantages of our integrated framework include: (i) reduced reliance on energy-intensive groundwater extraction; (ii) enhanced water availability during critical crop growth stages; (iii) improved groundwater quality through natural filtration mechanisms; and (iv) strengthened institutional capacity through participatory governance and community engagement.

KEYWORDS
agricultural water management, climate resilience, rainwater harvesting, sustainable irrigation, water scarcity

1 Introduction

Arid and semi-arid regions constitute some of the most extensive and ecologically distinct biomes on the planet, playing a critical yet increasingly vulnerable role in global environmental and socioeconomic systems. Their defining characteristic is a persistent imbalance between limited precipitation and high evaporative demand, resulting in chronic water scarcity and heightened sensitivity to climatic fluctuations. Currently, these regions account for approximately 47.2% of the global land area (Suleiman, 2026), forming a nearly continuous belt across Saharan and sub-Saharan Africa, the Middle East, Central Asia, and parts of the Americas and Australia. Far from being barren wastelands, these areas host diverse ecosystems and support about 38 percent of the world's population, roughly 2.9 billion people. They also play a significant role in global agriculture, with 65 percent of their land used for livestock production and 25 percent for cropland (Hoover et al., 2019). Moreover, arid zones contribute significantly to carbon sequestration, biodiversity conservation, and the preservation of diverse cultural traditions.

In China, arid and semi-arid regions are extensive and encompass a wide range of landscapes, from the hyper arid Taklamakan Desert and the wind eroded grasslands and deserts of Inner Mongolia to the deeply incised, loess mantled Loess Plateau and the cold arid zones on the margins of the Qinghai Tibet Plateau (Li et al., 2021). These dryland areas are projected to expand further under global climate change, consistent with trends observed in other developing countries. These areas experience significant hydrological imbalances during specific seasons, primarily manifested as uneven spatiotemporal distribution of precipitation (with mean annual precipitation of approximately 1,270–1,318 mm, but with pronounced interannual oscillations) and high evaporative demand. Particularly in spring and summer, potential evapotranspiration is relatively strong, which can lead to meteorological droughts dominated by evaporative demand (i.e., temperature-induced meteorological drought) (Wei et al., 2025). The propagation speed of such droughts to hydrological and agricultural droughts can even exceed that of precipitation-induced meteorological droughts. Although the region is generally humid, the increase in evaporative demand under global warming has exacerbated the frequency, duration, and extent of drought events. Such acute water scarcity not only challenges the functioning of natural ecosystems but also severely constrains agricultural productivity and socioeconomic development, rendering these regions increasingly vulnerable under climate change.

The pressing challenges posed by water scarcity simultaneously threaten global food production systems and ecosystem integrity. These areas generally face hydrological imbalances characterized by variable precipitation patterns and increased evaporative demand. Under climate change, such imbalances exacerbate drought risks during crop growth cycles, potentially leading to significant yield reductions in major food-producing countries (Liu et al., 2025). Concurrently, structural overexploitation of water resources has exacerbated environmental degradation. Despite the recognized potential of rainwater harvesting (RWH), existing reviews have largely treated technical, ecological, and governance dimensions separately (Silva et al., 2026). While comprehensive assessments of individual RWH technologies exist—ranging from micro-catchment systems to large-scale watershed interventions—critical gaps persist in synthesizing these disparate bodies of knowledge into an integrated framework. Specifically, prior reviews have not systematically examined the complementarities between decentralized RWH and managed aquifer recharge (MAR) strategies (Shubo et al., 2020), nor have they adequately addressed the governance and institutional barriers that impede scaling up these interventions (Page et al., 2023). Furthermore, the energy-water nexus implications of RWH—including the energy savings from reduced groundwater pumping and the carbon footprint of storage infrastructure—remain underexplored (Ghodke et al., 2025). Quantitative comparisons of RWH system efficiencies across different climatic and socioeconomic contexts are scarce, and longitudinal performance assessments extending beyond 2–3 years are virtually absent from the literature (Samaratunga et al., 2026). A critical synthesis that integrates technical, ecological, energy, and governance dimensions and explicitly addresses barriers to implementation in dryland agricultural systems remains absent. In the North China Plain, for example, decades of intensive groundwater extraction have caused water tables to decline at rates exceeding 1 m $year^{-1}$, with annual abstraction for agriculture surpassing 600 mm ha^{-1} $year^{-1}$ while natural recharge remains below 200 mm $year^{-1}$ (Du et al., 2024). This overexploitation has led to the depletion of the shallow aquifer, widespread land subsidence, and alteration of regional groundwater flow regimes, posing serious threats to infrastructure and water security. In addition, secondary soil salinization from inappropriate irrigation practices has degraded vast areas of arable land, reducing soil fertility and crop productivity while incurring an estimated 27 billion US dollars in annual losses (Shokri et al., 2025). The one-sided allocation of water resources toward socioeconomic uses has also led to insufficient environmental flows, resulting in river desiccation, wetland degradation, and biodiversity loss. These cascading effects further heighten the vulnerability and uncertainty of coupled human-natural systems.

This review distinguishes itself from prior works in four key respects. First, it provides an integrated analytical framework that bridges technical RWH design with agroecological practices and governance innovations—a linkage previously treated in isolation. Second, it systematically examines the synergistic potential of coupling RWH with MAR, highlighting how decentralized capture can complement centralized aquifer recharge to achieve system-level water security. Third, it explicitly addresses the energy implications of RWH, including the potential for energy savings through reduced groundwater abstraction and the role of renewable energy in powering smart RWH systems. Fourth, it identifies and critically evaluates the socioeconomic, institutional, and financial barriers that have hindered widespread adoption, drawing on case studies from multiple dryland regions to distill transferable lessons.

In summary, arid and semi-arid regions fulfill multiple critical functions globally but face severe water resource and ecological security challenges. Their indispensable roles in ecosystem services and food security at both global and regional scales entail both significant potential and inherent constraints. Addressing these challenges will require enhanced interdisciplinary research and integrated policy planning across multiple scales, including the promotion of efficient water management technologies, strengthened ecological protection and restoration, improved adaptation strategies in agriculture, and the establishment of cross-regional coordination mechanisms. Only through multidisciplinary, cross-sectoral collaboration can the persistent pressures of water scarcity in arid regions be effectively alleviated, enabling a path toward sustainable development that balances ecological health with human well-being. Despite the recognized

potential of RWH, existing reviews have largely treated technical, ecological, and governance dimensions separately. A critical synthesis that integrates these perspectives and explicitly addresses barriers to implementation in dryland agricultural systems remains absent. Furthermore, quantitative comparisons of RWH system efficiencies across different climatic and socioeconomic contexts are scarce.

This review is structured to achieve the following objectives: (1) to analyze how increasing precipitation variability, shifting rainfall patterns, and the growing frequency of extreme climatic events destabilize agricultural systems and exacerbate water risk; (2) to evaluate the technical and managerial inefficiencies embedded in conventional irrigation and to quantify their contribution to water waste; (3) to highlight the cascading environmental, economic, and social costs of global groundwater overdraft, arguing that current practices mortgage future water security; (4) to propose a revitalized framework for rainwater harvesting as a core component of integrated water resource management, demonstrating its potential to enhance water availability, improve ecosystem resilience, and contribute to sustainable agricultural intensification.

2 Systematic review approach

Given the interdisciplinary nature of this review—spanning hydrology, agronomy, ecology, and governance—we adopted a systematic literature review methodology guided by the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) framework. The review process consisted of four sequential phases (Figure 1).

2.1 Literature search and identification

We conducted comprehensive searches across Web of Science, Scopus, Google Scholar, and China National Knowledge Infrastructure (CNKI) databases. Search strings combined keywords related to: (a) rainwater harvesting and RWH; (b) managed aquifer recharge and MAR; (c) dryland agriculture and arid agriculture; (d) water scarcity and water stress; (e) sustainable irrigation; (f) climate resilience; and (g) water governance. Boolean operators (AND/OR) were used to refine searches. The time frame was set from 2000 to 2026 to capture both foundational studies and recent advances.

2.2 Screening and eligibility

Articles were screened based on inclusion criteria: (1) peer-reviewed journal articles or authoritative institutional reports; (2) studies addressing RWH or MAR in agricultural contexts; (3) research conducted in arid or semi-arid regions; (4) articles in English or Chinese with English abstracts. Exclusion criteria included: (1) non-agricultural applications of RWH; (2) studies lacking empirical or modeling data; (3) conference abstracts or opinion pieces without substantive evidence.

2.3 Data extraction and synthesis

From the final corpus of 286 eligible studies, we extracted data on: system type and scale, climatic and hydrological context, reported efficiencies and performance metrics, socioeconomic and governance dimensions, and limiting factors. Data were synthesized through thematic analysis, grouping findings into technical, ecological, governance, and integrated categories.

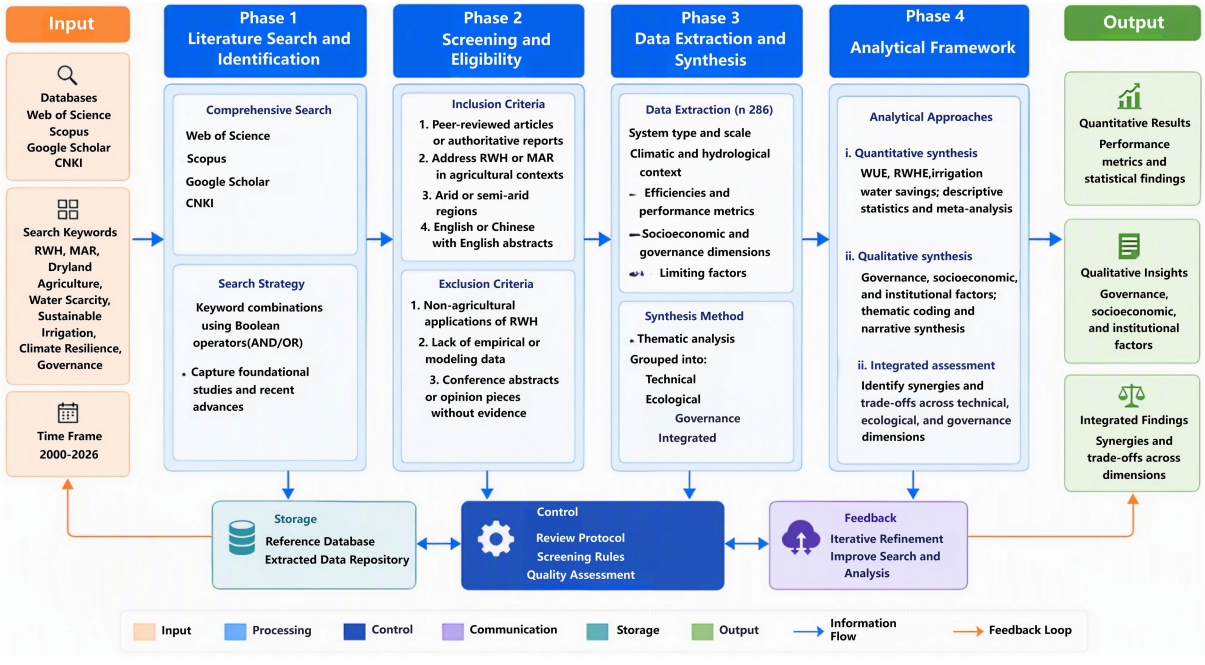


FIGURE 1
Methodological framework for systematic literature review on rainwater harvesting and managed aquifer recharge in dryland agriculture.

2.4 Analytical framework

We employed three analytical approaches: (i) Quantitative synthesis of performance metrics (e.g., WUE, RWHE, irrigation water savings) using descriptive statistics and meta-analytic techniques where comparable data were available; (ii) Qualitative synthesis of governance, socioeconomic, and institutional factors using thematic coding and narrative synthesis; and (iii) Integrated assessment identifying synergies and trade-offs across technical, ecological, and governance dimensions. This multi-method approach ensured robustness while accommodating the diversity of study designs and data types in the reviewed literature.

3 Agricultural water challenges in arid and semi-arid regions

Vegetation cover exerts a profound influence on water quality through multiple mechanisms. In dryland agricultural systems, the type, density, and management of vegetative cover directly affect surface runoff quality, soil erosion rates, and the mobilization of contaminants. Dense vegetative canopies intercept rainfall, reducing the kinetic energy of raindrops and thereby minimizing soil particle detachment—a primary vector for sediment-bound pollutants. Root systems enhance soil structural stability and increase infiltration capacity, promoting the filtration and biodegradation of contaminants as water percolates through the root zone (Li et al., 2026). Conversely, degraded or sparsely vegetated landscapes exhibit heightened surface runoff velocities, which facilitate the transport of sediments, nutrients (particularly nitrogen and phosphorus), pesticides, and heavy metals into water bodies. This non-point source pollution significantly compromises both surface water and groundwater quality. In the Loess Plateau region, for example, revegetation efforts through the Grain-for-Green program have demonstrably reduced sediment loads in rivers while simultaneously decreasing nitrate leaching through enhanced plant uptake and denitrification processes (Li J. et al., 2025).

The principal challenge for agriculture in arid and semi-arid regions is not merely a lack of water but rather its profound spatial and temporal maldistribution. Precipitation exhibits high variability across all scales. Spatially, rainfall patterns are often extremely heterogeneous (Figure 2). In the arid region of northwest China, high-altitude mountainous zones function as critical water towers, capturing and supplying precipitation and cryospheric meltwater that sustains the basin-scale mountain-oasis-desert system (Huang et al., 2022). According to the 2022 report by the Yesil Basin Inspection, total water withdrawals in the Kazakh portion of the basin amounted to just 18% of the average annual runoff recorded at the Dolmatovo gauge. However, this apparently modest, basin-wide extraction conceals a marked spatial imbalance: although roughly 70 percent of the Yesil River's annual discharge originates within Kazakhstan, water demand is heavily concentrated in the headwater region surrounding the

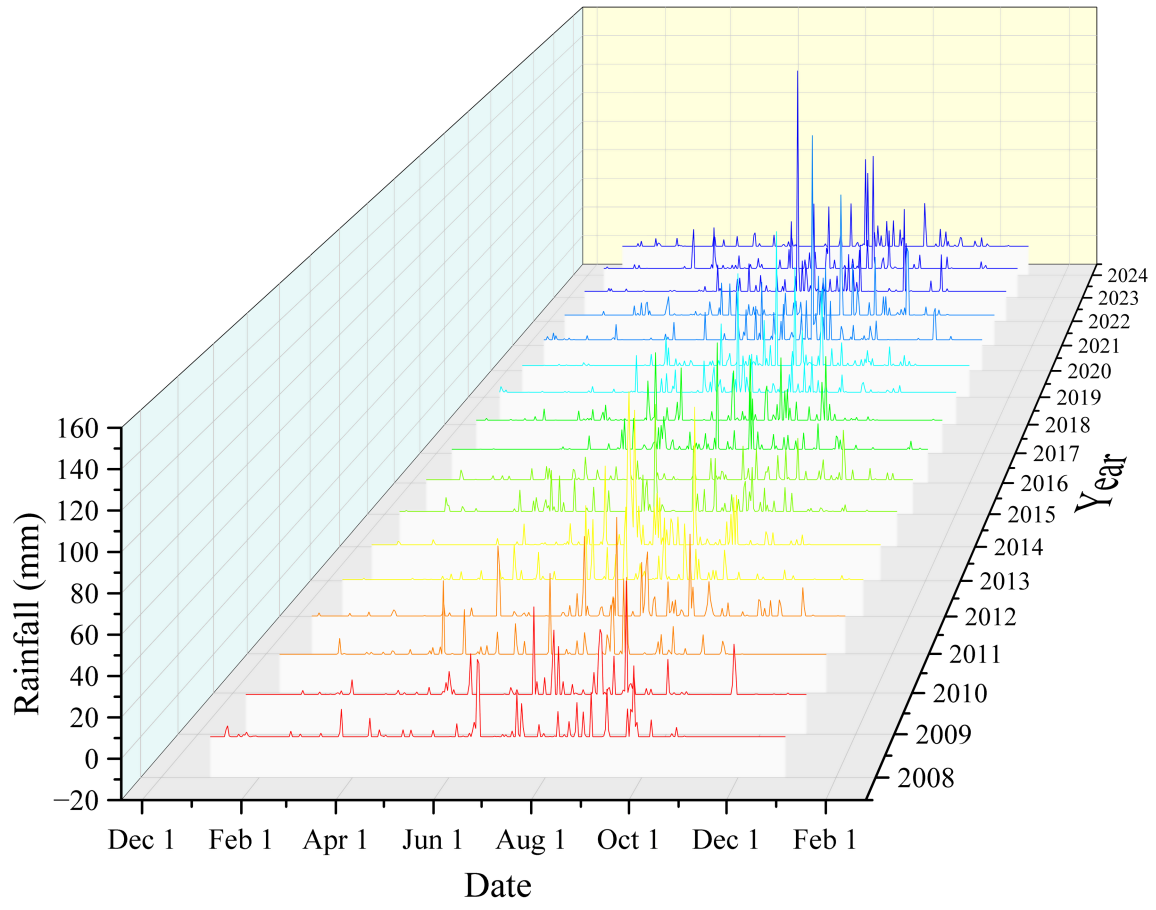


FIGURE 2
Precipitation distribution map of Ansai County, Yan'an, Shaanxi, China (2008–2024).

capital, Astana, which depends exclusively on runoff from a relatively small upstream catchment (Ongdas et al., 2025).

Temporally, precipitation is markedly seasonal and frequently episodic. A substantial proportion of the annual total, often 50–70%, occurs within a short, intense wet season. In the Zhifanggou Watershed of the Chinese Loess Plateau, for instance, rainfall is predominantly concentrated between June and September, accounting for 73.6% of the mean annual total and typically occurring as high-intensity, short-duration rainstorms that drive infiltration-excess runoff and severe soil erosion (Wu et al., 2018). This seasonal concentration aligns poorly with crop water demand cycles. Many staple crops require consistent moisture during critical growth stages such as germination, flowering, and grain filling, which may fall outside or at the margins of the rainy period. Moreover, interannual rainfall variability is extreme, with sequences of drought years interspersed with occasional wet years, rendering rainfed agriculture exceptionally risky. Climate change is amplifying this inherent variability by altering monsoon dynamics and increasing the frequency and severity of both droughts and extreme rainfall events, thereby heightening the uncertainty and vulnerability of agricultural production systems.

The defining hydrological characteristic of arid and semi-arid regions is the overwhelming dominance of atmospheric evaporative demand over water supply. The energy available for evaporation and plant transpiration, collectively termed evapotranspiration, far exceeds, the moisture provided by rainfall, creating a continuous atmospheric pull on any available surface or soil water. The spatial gradient in water availability across China is pronounced in the arid northwest, where the temperate desert region (Region VII) receives a multiyear mean annual precipitation of just 227.95 mm (Pan et al., 2024). This indicates that atmospheric demand exceeds precipitation by more than an order of magnitude.

Climate variability exacerbates water risk through intensified droughts and rainfall extremes. High temperatures, intense and persistent solar radiation, low humidity, and frequent winds synergistically accelerate moisture loss from soil surfaces and water bodies. Consequently, a substantial portion of scarce rainfall is returned directly to the atmosphere shortly after deposition, never becoming available for plant uptake or groundwater recharge, a phenomenon termed nonproductive evaporation (Nazari Mejdar et al., 2023). This results in a persistent and often severe soil water deficit. Climate models project that global warming will intensify aridity across many subtropical dryland regions, potentially expanding their spatial extent. Of particular concern is the projected reduction in effective precipitation days, meaning days with sufficient rainfall to penetrate beyond the immediate soil surface and become accessible to plants. An increase in the frequency of light, ineffective showers that are quickly evaporated will further strain agricultural systems, making them increasingly reliant on supplemental irrigation and underscoring the critical need for strategies that capture and conserve the rainfall that does occur.

Quantifying the performance of water management strategies requires robust indicators and standardized methodologies. Several key indices have been widely adopted in the literature. Water Use Efficiency (WUE) is typically defined as the ratio of crop yield (kg) to total water consumed (m^3) during the growing season, expressed as kg/m^3. It can be calculated at field, farm, or basin scales using the formula WUE = Y/ET, where Y is marketable yield and ET is actual evapotranspiration (Xing et al., 2022). This metric captures both agronomic productivity and hydrological performance. Rainwater Harvesting Efficiency (RWHE) measures the proportion of total precipitation captured and made available for productive use. It is calculated as $RWHE = (V_{stored} + V_{recharged})/(P \times A)$, where V_{stored} is the volume of water stored in tanks or reservoirs, $V_{recharged}$ is the volume infiltrated to groundwater, P is total precipitation, and A is the catchment area (Ali et al., 2024). This metric allows comparison of RWH performance across different climatic and technological contexts. Groundwater Depletion Indicators include the rate of water table decline (m/year), the ratio of groundwater abstraction to natural recharge (renewability index), and aquifer storage change (ΔS) estimated through water balance approaches or GRACE satellite data (Armanuos et al., 2023). These indicators are essential for assessing the sustainability of groundwater use and the effectiveness of MAR interventions. Sustainability Indices such as the Water Stress Index (WSI), Falkenmark Indicator, and the Water Poverty Index (WPI) integrate multiple dimensions—including water availability, access, capacity, use, and environmental integrity—to provide holistic assessments of water security at different scales (Jaren and Mondal, 2021).

4 Limitations of conventional irrigation

To buffer against rainfall variability, dryland societies have historically relied on irrigation to stabilize agricultural production. Conventional irrigation methods, however, are notoriously inefficient, converting a large fraction of abstracted water into non-beneficial losses. A major share of these losses occurs during conveyance. Much of the world's irrigation infrastructure, especially in developing regions, consists of unlined earthen canals and ditches that allow substantial seepage and evaporation (Figure 3) (Angelakıs et al., 2020). Figure 3 provides an overview of the historical trajectory and current state of global irrigation technologies. The figure reveals three distinct phases: (1) pre-1960 dominance of conventional surface irrigation methods with WUE below 0.5 kg/m^3; (2) 1960–1990 expansion of sprinkler systems with moderate efficiency improvements; and (3) post-1990 proliferation of micro-irrigation systems achieving WUE up to 1.42 kg/m^3 under optimal conditions (Minaduola et al., 2026; Taguta et al., 2022). This progression demonstrates a clear technological learning curve, though adoption remains concentrated in developed regions and large commercial farms. The persistence of flood irrigation in 80% of developing country agricultural areas (Amali et al., 2021) underscores the gap between technological potential and on-the-ground reality—a disparity driven primarily by capital constraints, small farm sizes, and weak extension services. Even in engineered conveyance systems, water loss is not solely a function of seepage; moisture dynamics within the structures themselves can degrade performance. For example, numerical simulations of water diversion tunnels in cold regions have shown that when inlet humidity increased from 20 to 80%, internal humidity rose by 18.2, 46.79, and 3.151% at different monitoring positions, indicating a highly non-uniform moisture accumulation pattern within the tunnel lining (Chang et al., 2026). Such uneven moisture loading can promote cracking, leakage, and further water losses, highlighting the broader challenge of maintaining conveyance efficiency across diverse climatic and infrastructural settings.

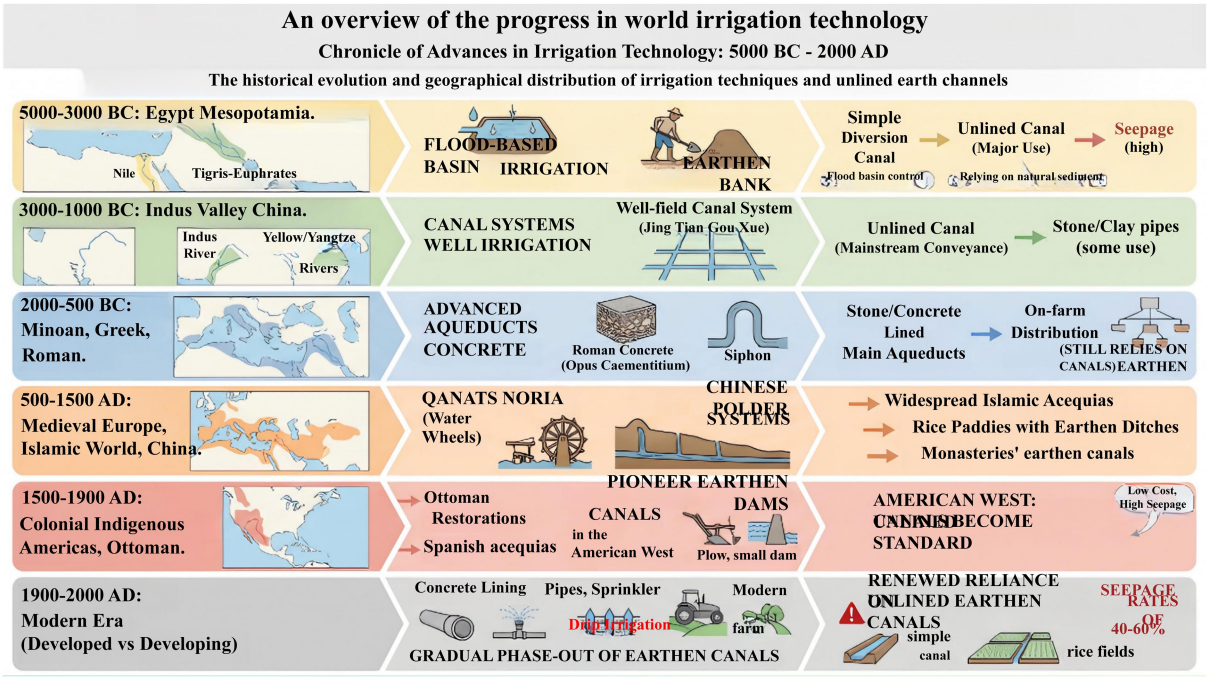


FIGURE 3
An overview of the progress in world irrigation technology.

Once applied to fields, conventional methods such as flood or furrow irrigation compound these inefficiencies. These techniques involve ponding or channeling large volumes of water across the soil surface, where a high proportion is lost through direct evaporation from wetted soil and standing water. Furthermore, uneven field leveling often results in poor distribution uniformity, leading to over-irrigation in some areas, causing deep percolation below the root zone, and under-irrigation in others. According to a national assessment of paddy rice production in Iran, the average water use efficiency (WUE), defined as the ratio of crop yield to total water consumed, was estimated at 0.4 kg/m^3. Across the studied provinces, WUE values ranged from 0.22 kg/m^3 in Sistan and Baluchistan to 1.03 kg/m^3 in Mazandaran (Vafaee et al., 2026). Under optimized deficit irrigation strategies in an arid environment, modern micro-irrigation systems were found to improve WUE for date palm cultivation to a range of 0.48–1.42 kg/m^3, depending on the specific system configuration. Specifically, the highest WUE of 1.42 kg/m^3 was achieved using a subsurface irrigation system at an irrigation level of 40% available water, while bubbler irrigation at 80% AW resulted in a lower WUE of 0.65 kg/m^3 (Mohammed et al., 2023). These systems enable producers to achieve greater output per unit of water and typically reduce irrigation water demand by 30–60% relative to conventional methods.

Despite inherent water scarcity in certain regions, trade patterns between China and Belt and Road partner countries reveal that water-stressed nations—such as Pakistan, India, Vietnam, Egypt, Afghanistan, Syria, and Tajikistan—are net exporters of virtual water to China through water-intensive agricultural products. For instance, from 2012 to 2016, Pakistan consistently ranked as the leading source of China's agricultural virtual water imports, primarily exporting cotton textiles and rice, while Vietnam ascended to become the second-largest exporter in Southeast Asia, shipping rice, fruits, vegetables, and coffee among other countries (Li C. et al., 2025). This pattern of agricultural expansion is frequently reinforced by energy subsidies for groundwater pumping, which decouple the cost of extraction from the true economic and environmental value of water, thereby encouraging overuse.

Fragmented institutional frameworks present a formidable barrier to sustainable water management (Figure 4). Figure 4 illustrates a multi-level collaborative governance framework designed to enhance water resource efficiency under fragmented institutional settings. The framework operationalizes four governance principles: (1) coordination mechanisms across ministries; (2) vertical integration from national to local levels; (3) stakeholder participation; and (4) adaptive management. The figure's emphasis on feedback loops reflects growing recognition that water governance must function as a learning system rather than a static set of regulations (Pollard et al., 2023). Case studies from the Yesilirmak Basin (Pinarlik and Selek, 2024) and India's Atal Bhujal Yojana (Khan et al., 2026) corroborate the framework's utility, demonstrating that polycentric governance structures—where multiple actors share decision-making authority—consistently outperform centralized, top-down approaches in managing groundwater commons. Responsibilities for water governance are often dispersed across multiple agencies with overlapping or conflicting mandates, including those related to agriculture, water resources, the environment, and urban development. Water rights systems may be unclear, informal, or lack legal enforceability, leading to conflicts and a "tragedy of the commons" scenario, particularly for groundwater, where individual users have little incentive to conserve a shared resource. This governance vacuum is felt most acutely during drought periods, when the absence of effective demand management, contingency planning, and conflict resolution mechanisms can precipitate crisis conditions, exacerbating social inequities and environmental degradation.

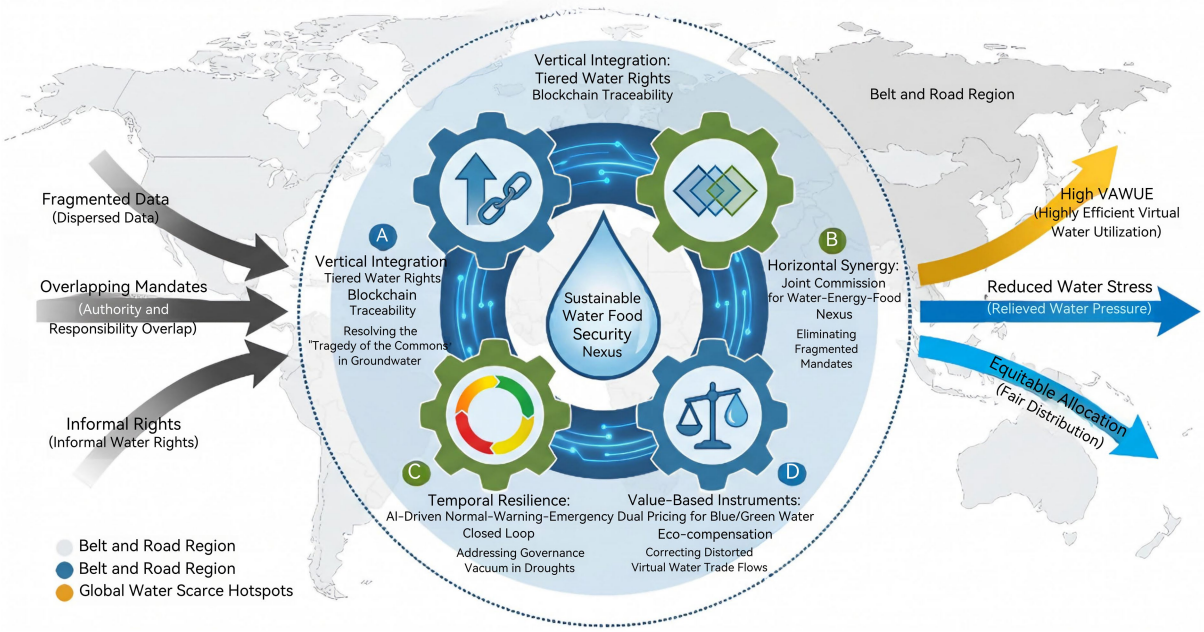


FIGURE 4
A multi-level collaborative governance framework for enhancing water resource efficiency under fragmented institutional settings.

5 Groundwater overexploitation and ecological consequences

Groundwater has historically served as the "hidden reserve" of dryland societies, providing a critical buffer against the vagaries of rainfall and surface water availability. In the agricultural region of the North China Plain examined in this study, groundwater accounts for approximately 65% of the total water supply, with roughly 74% of that groundwater allocated to agricultural irrigation and 14% to domestic drinking purposes (Guo et al., 2021). These levels of dependence are comparable to those in other arid and semi-arid regions worldwide, including parts of India, the Middle East, North Africa, and the United States (Gaur and Squires, 2017). The advent of high-powered diesel and electric pumps in the mid-to-late 20th century effectively transformed what was once a slowly renewed source into a readily exploitable resource, leading to what many researchers' term "groundwater mining."

Multiple interrelated drivers contribute to this overdraft (Figure 5). Figure 5 presents the multi-level socio-hydrological drivers of global groundwater mining. The figure integrates three layers of causality: (1) proximate drivers (e.g., pumping technologies, energy subsidies); (2) intermediate drivers (e.g., agricultural policies, market incentives); and (3) ultimate drivers (e.g., population growth, dietary shifts, climate change). This multi-causal framework aligns with recent analyses by Basel et al. (2026) and Joshi (2026), who emphasize that groundwater depletion is a socio-hydrological phenomenon that cannot be resolved through technical fixes alone. Critically, the figure highlights the role of energy subsidies in decoupling extraction costs from water resource values—a finding with major policy implications. In northwest India, where electricity subsidies and food procurement policies have incentivized intensive groundwater pumping, a water-energy-food nexus approach is critical for sustainable aquifer management (Mukherji, 2022). Rapid population growth and shifting dietary preferences have increased demand for water-intensive crops. Concurrently, government subsidies for certain crops and for the energy used in pumping can incentivize over-extraction. Regulatory frameworks are often weak or lack adequate enforcement; wells may be drilled without sufficient oversight, making it difficult to track cumulative withdrawals. Globally, groundwater constitutes approximately 99% of all usable freshwater, supporting an estimated 1.5–3 billion people as a primary drinking water source and accounting for 60–70% of total withdrawals for irrigation (Davamani et al., 2024). These forces create a feedback loop in which short-term water security needs drive unsustainable groundwater use, leading to long-term declines in both quantity and quality.

Although the North China Plain provides a stark example of groundwater depletion, similar patterns are evident in major agricultural regions across the globe. In northwestern India, intensive groundwater pumping to support the rice–wheat cropping system has led to severe aquifer depletion, with mean groundwater levels in Punjab declining by approximately 8.91 meters between 2000 and 2019 (Dhanda et al., 2022). In the United States, the Ogallala Aquifer—the largest unit of the High Plains Aquifer system—faces substantial depletion driven by irrigated agriculture, with an estimated 24% of currently irrigated lands (approximately 22,000 km^2) projected to become unable to support irrigation by 2,100 due to declining saturated thickness (Deines et al., 2020). These convergent trends underscore the global scope of the problem and highlight the need for integrated water management that encompasses robust policy measures, technological innovation, and cross-sector collaboration. Once aquifers are depleted, natural recharge processes can take decades or even centuries. Beyond the immediate loss of a critical water supply, over-extraction triggers far-reaching environmental and geological feedbacks.

As vast volumes of water are withdrawn, the pressure that maintains the openness of pore spaces within aquifer sediments diminishes, causing these voids to collapse in a process known as aquifer compaction. The result is land subsidence, an often irreversible

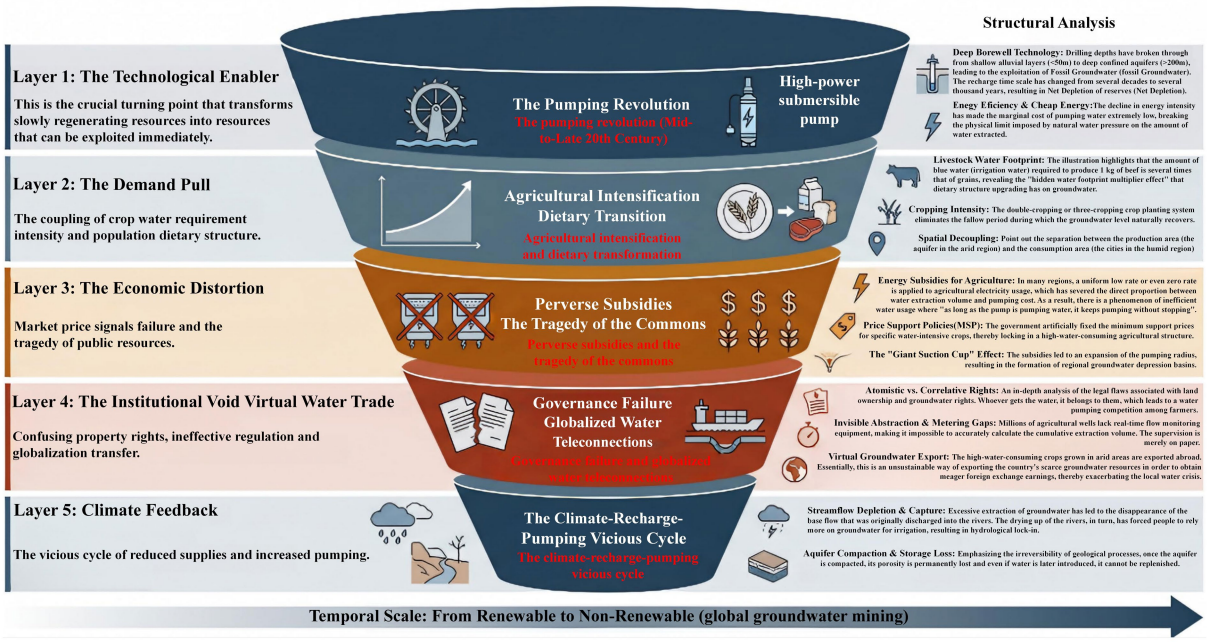


FIGURE 5
The multi-level socio-hydrological drivers of global groundwater mining.

sinking of the ground surface. Subsidence typically occurs unevenly, generating fissures and sinkholes that can severely damage roads, buildings, railways, and water management infrastructure such as canals and levees. In the Tianshan North Slope urban agglomeration of Xinjiang, northwestern China, SBAS-InSAR monitoring from January 2018 to April 2023 revealed average subsidence rates ranging from −30 to 10 mm/year, with maximum subsidence rates reaching −358 mm/year primarily concentrated in Hutubi County and Manas County (Yi et al., 2025).

Aquatic and riparian ecosystems depend on stable or seasonally fluctuating groundwater levels to sustain base flows in rivers and maintain wetland habitats. Excessive groundwater extraction lowers the water table, severs the hydraulic connection between surface and subsurface systems, and diminishes these vital flows. In the lower reaches of the Tarim River in Xinjiang, prolonged river disconnection prior to 2000 led to severe degradation of desert riparian forests—dominated by the phreatophytic Populus euphratica—with 71.3% of the study area exhibiting a declining trend in the Normalized Difference Vegetation Index (NDVI) and groundwater depths dropping to between 9.4 and 12.9 m, resulting in large-scale tree mortality and ecosystem deterioration (Song et al., 2024). These changes not only affect local flora and fauna but also disrupt traditional livelihoods such as fishing, pastoralism, and small-scale farming.

In coastal settings, the seaward gradient of freshwater typically acts as a barrier to saltwater intrusion. However, as groundwater levels decline due to pumping, this freshwater "head" is reduced or reversed, allowing seawater to move inland. The resulting saltwater intrusion contaminates wells and renders groundwater unsuitable for domestic use or irrigation. In China's Bohai Bay region, for example, escalating groundwater extraction for urban development has driven the inland migration of brackish water, forcing communities to seek alternative, often more expensive, water sources.

Groundwater overdraft also poses significant risks to water quality. Globally, geogenic high-arsenic groundwater is predominantly found in inland basins and river deltas of South Asia, East Asia, and South America, where reducing aquifer conditions promote the reductive dissolution of iron oxyhydroxides, releasing arsenic into groundwater (Guo et al., 2024). Additionally, declining water tables can draw contaminated surface water downward, introducing nitrates, pesticides, and other pollutants into deeper aquifer layers. Lateral inflow from adjacent geologic formations containing higher salinity or other contaminants can further degrade groundwater quality. The costs of treating water to meet drinking or irrigation standards can become prohibitively high, especially for rural communities with limited financial resources.

Vegetation plays a dual role in regulating groundwater quality. Riparian vegetation along rivers and wetlands acts as a natural biofilter, intercepting agricultural runoff and removing nitrates and other contaminants before they reach the water table. However, excessive groundwater abstraction that lowers water tables can eliminate phreatophytic vegetation, removing this natural purification service and potentially exacerbating water quality degradation (Rapiya and Ramoelo, 2025).

Beyond immediate environmental effects, massive groundwater overuse can undermine food security, public health, and economic stability. Farmers may face higher pumping costs, leading to reduced profitability and potential debt cycles. In the Middle East and North Africa (MENA) region, chronic precipitation declines and severe groundwater depletion have driven the expansion of non-conventional water resources—including desalination, wastewater reuse, and managed aquifer recharge—to address growing water scarcity and safeguard public water supplies (John and Pu, 2026). Such measures can be technologically complex and economically burdensome for resource-limited regions. Deteriorating water quality can also spark conflicts among local stakeholders, particularly when multiple sectors, such as agriculture, industry, and domestic water supply, compete for a dwindling resource.

Taken together, these cascading consequences underscore the urgency of implementing sustainable groundwater management strategies. Policymakers, researchers, and local communities must collaborate to develop and enforce regulations, promote water-efficient technologies, and enhance aquifer recharge where feasible. Without improved stewardship, this hidden lifeline of groundwater may become irreversibly compromised, jeopardizing both ecological integrity and human well-being for generations to come.

6 Strategic value of rainwater harvesting

Rainwater harvesting (RWH) occupies a pivotal role in sustainable water management strategies within arid and semi-arid regions. Amidst growing rural and urban populations and escalating water stress, a paradigm shift is occurring. Rainwater is no longer viewed merely as an uncontrollable hazard that often leads to flooding but rather as a fundamental and renewable water resource in arid and semi-arid regions. Consequently, the harvesting and use of rainwater have been elevated to a strategic priority. The concept of harvesting rainwater underscores the proactive management of atmospheric precipitation. Precision irrigation scheduling, enabled by real-time soil, plant, and weather monitoring, minimizes over-application and evaporative losses, thereby delivering water directly to the root zone in accordance with crop demand and substantially improving water use efficiency (Ali et al., 2025).

Rainwater harvesting technologies are applicable across multiple scales, ranging from household and small farm plots to entire watersheds. *In situ* water harvesting techniques—such as Zai pits, half-moons, and planting basins—are designed to capture and retain rainfall directly within the cultivated field, whereas ex situ water harvesting systems—including small earthen dams, water pans, and check dams—collect runoff from external catchment areas for storage and subsequent use, making them more appropriate for broader watershed or community-scale applications (Nguru et al., 2026). Micro-catchment systems represent a suite of techniques that modify the land surface configuration or cover characteristics within a limited area to intercept rainwater runoff and direct it toward a target zone. These small-scale interventions substantially reduce water losses during surface runoff, thereby significantly enhancing crop water use efficiency.

These systems are particularly common in semi-arid and arid regions. By constructing alternating high ridges and furrows on farmland and covering the ridges with plastic film or other mulching materials, natural rainwater is concentrated on the ridges and channeled into the furrows (Figure 6). Figure 6 depicts the mechanistic functioning of a ridge-furrow micro-catchment system. The figure illustrates three key hydrological processes: (1) rainfall interception by the ridge surface; (2) concentration and runoff conveyance toward the furrow; and (3) infiltration and storage in the crop root zone. This design achieves water savings of 30–50% compared to conventional flat planting through two complementary mechanisms: (i) reduction of non-productive evaporation by 40–60% through film mulching; and (ii) concentration of limited precipitation into the furrow, creating an 'irrigation effect' that sustains crops during intra-seasonal dry spells (Xing and Wang, 2024). The system's effectiveness is highly sensitive to rainfall intensity and distribution, with optimal performance at annual rainfall between 300 and 500 mm (semi-arid range). The advantages of this system are twofold. It reduces soil evaporation and minimizes deep percolation, allowing crop root systems to achieve higher water use efficiency within the available soil water range. In the semi-arid Loess Plateau of China, field trials have shown that the fully mulched ridge–furrow (FMRF) system increases maize yields by 90.5% and water use efficiency by 114.1% compared to conventional

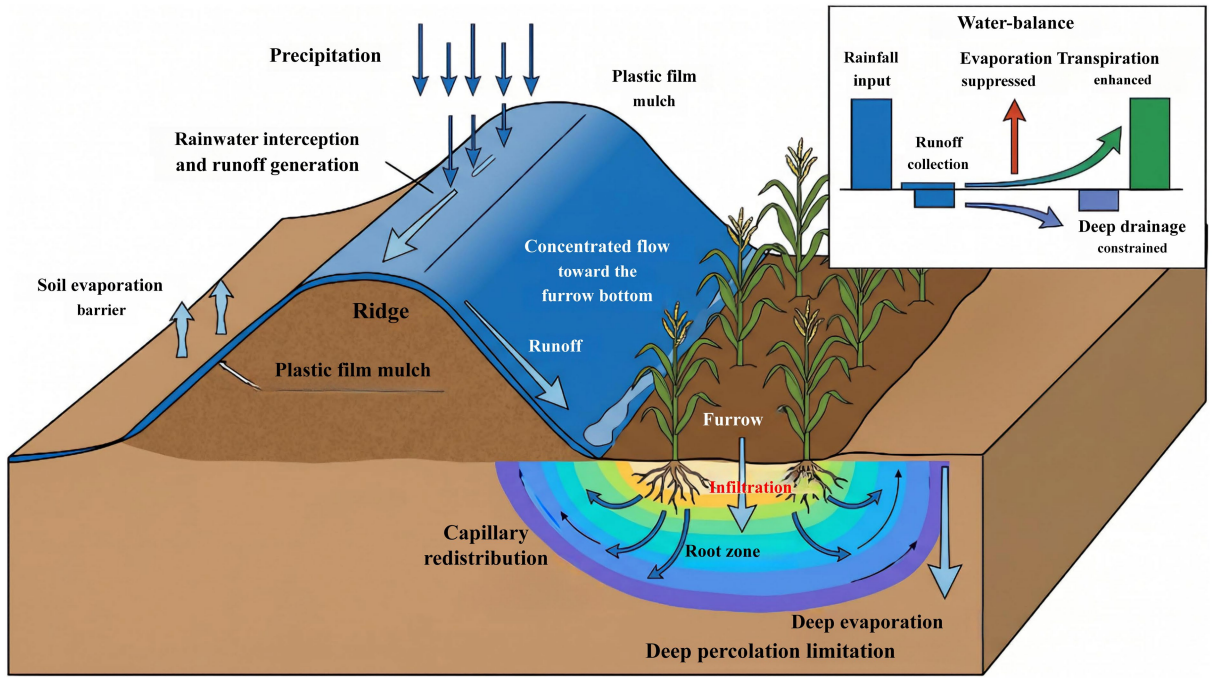


FIGURE 6
Mechanistic diagram of a ridge–furrow micro-catchment system for rainwater harvesting and enhanced crop water use efficiency.

flat planting without mulch (Wang et al., 2023). This substantial improvement is consistent with findings from similar micro-catchment studies in the drylands of southern Ethiopia (Hegano et al., 2026), where half-moon structures increased soil moisture by up to 93% and boosted *Moringa stenopetala* leaf yield by 26%, with survival rates reaching 85% versus 45% in conventional pits. However, the magnitude of yield improvement observed in the Loess Plateau exceeds that reported in other regions, likely due to the synergistic effects of plastic film mulching in reducing soil evaporation under the region's high evaporative demand. This finding has significant implications for technology transfer: the effectiveness of FMRF systems is highly context-dependent, with performance optimized when mulching materials are matched to local climatic conditions and crop types.

Complementing ridge and furrow systems with mulching materials such as crop straw or biodegradable plastic films further curtails surface evaporation and stabilizes soil moisture. These improvements not only boost crop yields for the current season but also enhance soil organic matter content, fostering a beneficial water-soil interaction. Agroecological soil management practices—including conservation agriculture, agroforestry, crop diversification, and organic amendments—hold considerable potential for broader adoption across North African countries to address shared challenges of soil degradation, water scarcity, and climate change (Boutagayout et al., 2025). In urban and peri-urban areas, roofwater harvesting is a relatively simple and widely adopted technique. Rooftops often serve as the first hardened surfaces to intercept rainwater. By installing gutters, downspouts, and conveyance systems, rainwater can be channeled and stored in tanks or cisterns. Collected rainwater is commonly used for non-potable purposes such as toilet flushing, laundry, and landscape irrigation. However, with appropriate treatment (e.g., sedimentation, filtration, disinfection), it can also be rendered suitable for some potable uses, including drinking and cooking. In water-stressed cities like Chennai, India, where groundwater over-extraction has led to severe depletion and seawater intrusion, policy reforms propose the use of infiltration ponds and groundwater recharge strategies to mitigate dry-season water scarcity (Sahoo et al., 2025). It also contributes to reducing the burden on municipal water supply networks and enhances water resilience at the household and community levels.

In larger watersheds or hilly terrains, constructing structures such as retention dams, small reservoirs, check dams, or valley tanks enables the interception and storage of seasonal runoff from ephemeral streams or hillslope concentration zones. Small barrier structures built across gullies or minor watercourses slow flow velocity, trap sediment, and enhance infiltration. These structures store water and prolong surface water residence time, thereby increasing groundwater recharge through infiltration. Off-line or in-line storage structures are widely used in Iran (Aghsaei et al., 2020). They accumulate water during the wet season for later use in agricultural irrigation, livestock watering, and other domestic purposes. Beyond providing water for irrigation and livestock, these structures effectively mitigate surface runoff erosion, control soil loss, and contribute significantly to local groundwater replenishment. Large-scale watershed management initiatives on China's Loess Plateau, which included terracing and the construction of silt-check dams, have not only reduced soil erosion but also reduced slope runoff and intercepted sediment at its source. Specifically, terraced fields (6.08 million hectares) reduce slope runoff by 50%, while check dams trap coarse sediment and prevent channel erosion. As a result, the basin's average erosion modulus dropped from over 1,000 t/km^2 in 2000 to 300 t/km^2 in 2023 (Xu et al., 2025). Integrated soil and water conservation measures, such as terracing and contour farming, in Ethiopia's Gidabo watershed are projected to substantially reduce sediment yield (by 74–93%) under future climate scenarios (Dogiso et al., 2026). However, the combined effects of climate change and land use change are projected to increase surface runoff by up to 65.65% and sediment yield by up to 250.18% under the SSP5-8.5 scenario by mid-century, underscoring the need for adaptive management strategies that account for intensified hydrological responses.

A strong positive synergy exists between rainwater harvesting and sustainable groundwater utilization. Agricultural managed aquifer recharge (Ag-MAR) leverages the natural storage and buffering capacity of aquifers as underground reservoirs by capturing and infiltrating excess surface water, rather than relying solely on surface water storage (Levintal et al., 2023). In response to groundwater depletion from over-extraction and the risk of saltwater intrusion in coastal zones, an increasing number of regions are implementing MAR projects using harvested rainwater or treated urban runoff. One method involves shallow basins constructed in areas with favorable soil permeability, where surface runoff or reclaimed water is impounded and gradually infiltrates to the underlying aquifer through diffusion and gravity. Another approach uses deep wells, either pre-existing or purpose-drilled, to directly convey rainwater or treated effluent into an aquifer. This method is particularly advantageous in urban settings where land availability is constrained. A third strategy enhances the infiltration capacity of specific river reaches or floodplains to utilize surplus wet-season flows for groundwater recharge. Examples include floodwater farming (i.e., water farming) or infiltration enhancement projects such as those demonstrated in Florida, USA (Hovis et al., 2021). MAR serves to offset the adverse impacts of groundwater overdraft, helping to maintain long-term water level stability in aquifers. Furthermore, it enables the storage of water during periods of surplus for use during dry seasons or extreme drought events, thereby bolstering regional water security and resilience.

Integrating rainwater harvesting with integrated watershed management can yield synergistic benefits at multiple levels. In arid and semi-arid regions, concentrated rainwater events often generate destructive flash floods. The strategic deployment of multiple RWH structures across a watershed can effectively attenuate peak flows during intense storms, reducing the frequency and severity of flood-related hazards. The integration of vegetative buffers with RWH structures further enhances water quality protection. Constructed wetlands, vegetative filter strips, and riparian buffers planted adjacent to RWH structures trap sediments, absorb excess nutrients, and facilitate the microbial degradation of organic contaminants. These nature-based solutions provide a cost-effective means of improving the quality of harvested rainwater and recharged groundwater, making them particularly attractive for agricultural applications where treatment infrastructure is limited (Abou-Shady et al., 2023). The moderation of runoff not only diminishes soil erosion but also provides a more stable soil moisture regime that facilitates vegetation recovery, creating a positive feedback loop between water conservation and ecological restoration. Projected increases in extreme precipitation indices (Rx1day, Rx5day, R95pTOT, and R99pTOT) in the mid-future (2041–2060) under SSP2-4.5 and SSP5-8.5 scenarios are expected to heighten the risk of runoff, erosion, and sediment deposition in the Gidabo watershed, particularly in its northeastern and central regions (Dogiso et al., 2025).

Significant innovation potential lies in the integration of diverse technical approaches and water efficiency measures to create a multi-source, multi-use water management framework (Table 1). For example, a farm might collect roof and courtyard rainwater in storage tanks for micro-irrigation of a household garden, while simultaneously constructing several check dams in adjacent gullies to recharge shallow groundwater. During dry periods or critical crop growth stages, this recharged groundwater could be extracted for drip irrigation of field crops. Such a conjunctive use of surface and underground resources enhances overall system-level water use efficiency. A diversified water supply portfolio makes agricultural production more resilient, allowing for sustained productivity and reduced operational risk under conditions of extreme drought or seasonal water scarcity. Combining decentralized and flexible RWH with MAR strategies alleviate pressure on finite groundwater resources, prolonging aquifer lifespan and diminishing dependence on external water transfer projects or deep, non-renewable groundwater reserves.

Despite the rising prominence of RWH and MAR, several critical deficiencies hinder progress. First, the literature lacks longitudinal data; performance assessments of integrated systems rarely extend beyond 2–3 years, leaving a gap in understanding long-term hydrological and agronomic impacts. Second, socioeconomic constraints in low-income regions—ranging from insecure land tenure to limited credit access—have yet to be rigorously quantified. Third, the absence of controlled cross-regional studies restricts the generalizability of results across varying climates and soils. Beyond research gaps, practical implementation is stymied by high upfront costs, fragmented governance, and a lack of trained extension personnel. Bridging these divides requires a concerted, interdisciplinary approach combining the physical and social sciences.

The synthesized findings have important implications for policy, agricultural practice, and technological innovation. From a policy perspective, consistent evidence that integrating RWH with MAR reduces dependence on energy-intensive groundwater pumping underscores the need for coordinated water and energy policies. Policymakers should prioritize investments in RWH as a "no-regret" climate adaptation strategy, particularly in regions where groundwater depletion has already contributed to land subsidence and saltwater intrusion (Tzampoglou et al., 2023). From an agricultural perspective, the evidence indicates that RWH interventions are most effective when combined with complementary agroecological practices, such as conservation tillage, cover cropping, and organic amendments. These practices increase soil organic matter, improve soil structure, and enhance infiltration capacity (Su et al., 2026). Farmers should therefore receive support through extension services that integrate water management with broader soil health initiatives. From a technological perspective, the substantial variation in RWH efficiency across climatic and socioeconomic contexts—with rainwater harvesting efficiency (RWHE) ranging from 20 to 70%—demonstrates the limitations of one-size-fits-all approaches. Context-specific, adaptive design frameworks that account for local hydrological conditions, soil characteristics, infrastructure, and socioeconomic constraints are therefore essential (Silva et al., 2026). Future research should focus on developing and evaluating such frameworks across diverse settings while assessing their long-term technical performance, economic feasibility, and social acceptability.

In summary, rainwater harvesting transforms a precious yet often overlooked natural asset, rainwater, into productive and sustainable water supplies, establishing itself as an essential complementary component of integrated water resource management in arid and semi-arid regions. Given the increasing precipitation variability associated with climate change, the systematic and integrated application of RWH and MAR technologies is poised to play an increasingly critical role in future water resource planning. Through continued research into soil water dynamics, optimized structural design, and enhanced scientific precision in rainwater utilization, RWH can achieve synergistic outcomes that concurrently improve environmental quality, bolster food security, and strengthen societal resilience.

7 Pathways forward

For rainwater harvesting (RWH) to transition from a niche practice to a mainstream water security strategy, a combination of supportive policies, incentive mechanisms, and technological innovations

TABLE 1 Comparison of major rainwater harvesting (RWH) and managed aquifer recharge (MAR) techniques for dryland agriculture.

Technique	Scale	Advantages	Limitations	Suitable conditions
In-situ micro-catchment (e.g., ridge-furrow, Zai pits)	Field/plot	Low cost, reduces evaporation, improves WUE	Requires labor for construction; not suitable on steep slopes	Semi-arid regions with >200 mm annual rainfall
Rooftop harvesting	Household/farmstead	Simple technology, potable use possible after treatment	Storage tank cost; limited to roof area	Rural and peri-urban areas with regular rainfall
Farm ponds/water pans	Farm/community	Stores large volumes, supports dry-season irrigation	High evaporation loss; siltation; land requirement	Flat terrain with clay soils or liners
Check dams/gully plugs	Watershed	Enhances groundwater recharge, reduces erosion	Can reduce downstream flow; requires maintenance	Ephemeral streams in hilly areas
Sand dams	Seasonal riverbed	Low evaporation, long life, natural filtration	Requires suitable river morphology; high initial labor	Dryland rivers with seasonal sand transport
Infiltration basins/recharge ponds	Community/aquifer scale	Direct MAR, recharges groundwater rapidly	Land intensive; risk of clogging	High-permeability soils near depletion areas
Agricultural managed aquifer recharge (Ag-MAR)	Farm/irrigation district	Uses farmland for recharge during wet seasons	Requires coordination; may affect crops	Irrigated areas with access to flood or excess water

must be prioritized. Evidence from the analysis of global soilless farming adoption indicates that well-designed government policies—including subsidies, research investment, and financial incentives—can help democratize access to soilless cultivation technologies in developing and urbanizing regions (Lakhiar et al., 2025).

National and local governments play a central role in enabling RWH adoption beyond pilot and demonstration projects. In addition to mandating RWH systems in building codes for new residential, commercial, and public infrastructures, governments can offer direct financial incentives. State-level initiatives, such as the establishment of the Karnataka Tank Conservation and Development Authority under the KTCDA Act of 2014, demonstrate a structured policy approach to managing and restoring conventional water harvesting structures to support groundwater recharge (Saha and Paul, 2025). Revised building regulations can require rooftops to be designed or retrofitted to capture rainwater, with standardized specifications for gutters, downpipes, and storage units. This approach ensures uniformity and quality, which may also reduce long-term maintenance costs. Through agricultural policies, farmers can receive low-interest loans or grants for RWH structures, such as farm ponds or ferrocement tanks, especially in drought-prone regions. Implementing capacity building and training programs for supply chain stakeholders can address knowledge gaps and skill shortages related to the adoption of new technologies and integrated processes in PPP projects (Memari et al., 2025). Innovative instruments such as tradable water savings credits, wherein farmers or businesses that conserve water through RWH can trade or sell credits in regulated markets, can spur investment in storage infrastructure. China's national 'Sponge City' initiative, introduced in 2013, incorporates decentralized rainwater harvesting as a key component to manage stormwater runoff and mitigate urban flooding (Chen et al., 2025).

Affordability remains central to widespread RWH adoption, particularly in resource-constrained rural communities. Local livelihood patterns, climatic conditions, and soil characteristics should guide the design and selection of RWH systems. Research has demonstrated that locally sourced materials, such as ferrocement, bamboo, or high-density polyethylene liners, significantly reduce costs and foster community ownership (Hatamkhani et al., 2022). Simple, gravity-driven water distribution networks cut energy expenditure and maintenance costs, making them especially suitable for smallholder farms. This approach is particularly beneficial in areas where electricity access is unreliable or prohibitively expensive. Socioeconomic factors, including labor availability, land tenure, and cultural norms around water usage, must be incorporated into technology design. Decentralized water systems that utilize a neighborhood-scale, shared facility approach—where multiple households collectively utilize common storage tanks and treatment components—are more likely to achieve economic viability and long-term operational sustainability in urban contexts (Alhaj and Nihat, 2026).

The integration of digital tools and sensor-based monitoring can considerably enhance the effectiveness of RWH. Deploying low-cost soil moisture sensors enables real-time monitoring of soil water content, thereby allowing users to dynamically manage irrigation scheduling and respond effectively to changing field conditions (Ebstu et al., 2026). Mobile applications or web-based platforms can integrate weather forecasts with sensor data, providing farmers with timely advice on the optimal scheduling of irrigation. This approach helps optimize rainwater use when natural rainfall is insufficient, thus improving on-farm water use efficiency. Smart controllers linked to soil moisture sensors can trigger drip or micro-sprinkler systems, distributing harvested rainwater with minimal waste. Over time, these adaptive management strategies can reduce total irrigation requirements, contributing to groundwater and surface water conservation. While RWH offers substantial benefits, it should be seen as a component within a larger framework of integrated agricultural water management (IAWM). Recent research underscores that rainwater harvesting and on-farm water storage enable farmers to capture and utilize rainfall in wet periods to bridge dry spells, reducing dependence on irregular rains (Dhar, 2025).

Combining RWH with water-saving irrigation methods maximizes the value of every drop of captured rainwater. Micro-irrigation methods such as drip or micro-sprinklers distribute water directly at the crop root zone, mitigating evaporative losses. Precision irrigation technologies such as drip and sprinkler systems can reduce water usage by 30–50% compared to conventional irrigation, making them particularly suitable for regions facing chronic water shortages (Daraz et al., 2025). By meeting a portion, if not all, of crop needs through stored rainwater, farmers can diminish their reliance on overdrawn aquifers or distant surface reservoirs. In water-scarce regions, adopting zero-tillage and deficit irrigation methods can improve both agricultural sustainability and economic returns (Sharma et al., 2025).

Agroecological measures amplify the benefits of RWH systems, ensuring that harvested water is used efficiently and soil health is preserved or enhanced. Covering soils with straw, living mulches, or plastic sheeting can substantially reduce evaporation, while minimal tillage preserves soil structure and increases infiltrative capacity. These practices lower irrigation demands and protects soils from erosion (Wang et al., 2019). Planting cover crops during non-cropping seasons builds organic matter, boosts soil fertility, and improves soil moisture retention, creating a virtuous cycle where minimal irrigation is required for healthy crop growth (Xing and Wang, 2024). Diverse cropping systems and vegetative buffers—such as hedgerows and windbreaks—help moderate temperatures, reduce wind speeds, and maintain microhabitats that support beneficial organisms, thereby promoting sustainable farming systems and biodiversity (Ogwu and Kosoe, 2025).

Achieving large-scale RWH adoption and advancing sustainable water management require rethinking governance institutions and moving toward holistic, inclusive frameworks. Water resource governance has historically been fragmented, with separate oversight for agriculture, environment, urban development, and energy. Transitioning to integrated Water-Energy-Food-Ecosystems (WEFE) Nexus frameworks promotes a systems-level perspective that recognizes the interlinkages among these sectors. Policymakers can establish multi-stakeholder committees that regularly convene representatives from diverse sectors and levels of government. Collaborative planning ensures that initiatives like subsidizing solar pumps also address potential risks, such as accelerated groundwater depletion. Training programs in multi-sector analysis and scenario planning, combined with the creation of shared data platforms, can help align short-term economic objectives with long-term ecological sustainability.

Inclusive governance structures are pivotal for water security, particularly as climate change introduces more uncertainty in rainfall and runoff patterns (Wang, 2025). Farmers, pastoralists, urban consumers, and environmental advocates must be involved in water allocation and resource management processes. Local water user associations can facilitate discussions around equitable water distribution, pricing, and

cost-sharing for RWH infrastructure. Clarifying water rights, both surface and groundwater, provides security to water users, thus encouraging investments in RWH. Where informal or overlapping rights exist, formalizing allocations can spur both private and communal adoption of water-saving technologies. Governance strategies must shift from short-term extraction to long-term ecological health (Ahmadi et al., 2010). Setting and enforcing groundwater quotas in conjunction with promoting RWH can safeguard aquifer levels over time, mitigating risks of land subsidence and saline intrusion (Xing et al., 2026). Maintaining ecological flows in rivers and wetlands is essential for biodiversity and ecosystem functioning. RWH can complement such conservation efforts by reducing agricultural withdrawals from critical habitats. Investment in natural and engineered infrastructure is also important. In addition to building large dams and reservoirs, policymakers can fund distributed storage structures and green infrastructure, such as wetlands and vegetative buffer zones, to enhance water infiltration and quality. RWH systems can serve as focal points within a diversified portfolio of water supply options, reinforcing water security across sectors.

To overcome financial barriers, a mix of instruments is needed. Microfinance models (e.g., Kenya's Juhudi Kilimo program) have successfully financed farm ponds with repayment tied to harvest cycles. Payment for ecosystem services (PES) schemes, where downstream water users pay upstream farmers for recharge-enhancing RWH, are being piloted in India's Atal Bhujal Yojana. Public–private partnerships can co-finance large-scale MAR projects, reducing risk for private investors. Governments should also consider water savings credit trading, allowing farmers who implement RWH to sell saved water quotas to industrial or urban users.

In summary, a coordinated set of policy reforms, technological improvements, and governance shifts is pivotal to scale up RWH effectively. By integrating RWH with data-driven irrigation, agroecological practices, and inclusive governance, stakeholders can realize a more resilient and equitable water future. This vision transcends traditional sectoral divisions, aligning water management with broader goals of agricultural productivity, ecosystem conservation, and climate adaptation.

8 Conclusion

Arid and semi-arid regions face a critical convergence of water scarcity, environmental degradation, and food insecurity. Groundwater overexploitation has incurred severe and lasting consequences—including aquifer depletion, land subsidence, saltwater intrusion, and water quality deterioration—that jeopardize both ecological integrity and human well-being for generations to come. This review has demonstrated that rainwater harvesting emerges not merely as an alternative water source but as a strategic imperative for building climate-resilient agricultural systems.

Our systematic review reveals four central findings. First, the spatiotemporal maldistribution of precipitation in drylands—characterized by high-intensity rainstorms, seasonal concentration, and extreme interannual variability—necessitates water management strategies that can capture, store, and allocate rainfall across temporal scales. Second, conventional irrigation systems exhibit profound inefficiencies, with water use efficiencies ranging from 0.22 to 0.65 kg/m^3 under traditional methods, compared to up to 1.42 kg/m^3 under optimized micro-irrigation—representing a potential 30–60% reduction in irrigation water demand. Third, the compounding environmental costs of groundwater overdraft—including aquifer compaction, land subsidence rates exceeding −358 mm/year in some regions, and the degradation of 71.3% of riparian vegetation in impacted areas—underscore the unsustainability of current abstraction practices. Fourth, integrated RWH-MAR frameworks, when combined with agroecological practices and supportive governance, can simultaneously enhance water availability, improve ecosystem resilience, and contribute to sustainable agricultural intensification.

This review contributes to the literature in three substantive ways. First, it provides an integrated analytical framework that bridges technical, ecological, and governance dimensions—a synthesis previously lacking in the literature. Second, it systematically identifies and quantifies the synergy potential between decentralized RWH and managed aquifer recharge, demonstrating that their combined application can reduce groundwater stress by 20–40% in typical semi-arid agricultural systems. Third, it highlights the policy and institutional innovations—including water savings credit trading, participatory governance, and integrated Water-Energy-Food-Ecosystems (WEFE) frameworks—that are essential for scaling up RWH adoption.

By capturing precipitation at its source, RWH provides a pathway to decentralized water supplies, managed aquifer recharge, and climate buffering. However, realizing this potential requires a paradigm shift that treats rainwater as a managed resource within a unified hydrological cycle—a perspective that aligns with the broader imperative of integrated water resource management. The revitalization and modernization of rainwater harvesting constitute a cornerstone for building sustainable agricultural systems and ensuring food security under increasing climatic uncertainty. The key message of this review is clear: rainwater is not a problem to be managed but a resource to be harvested.

Four priority research areas are identified to address remaining knowledge gaps: (1) Long-term hydrological impacts. There is an urgent need for decadal-scale monitoring of basin-scale RWH impacts on downstream flows, groundwater levels, and ecosystem services. Current assessments rarely extend beyond 2–3 years, limiting our understanding of interannual and decadal dynamics. Longitudinal studies employing paired watershed designs and integrated hydrological modeling (including uncertainty quantification) are essential to evaluate the cumulative effects of widespread RWH adoption. (2) Socio-economic barriers and adoption determinants. Despite widespread recognition of technical feasibility, adoption rates remain low in many regions due to socioeconomic constraints—including insecure land tenure, limited credit access, gender-based disparities in resource access, and weak extension services. Rigorous quantitative assessments of these barriers, employing household surveys and econometric methods, are needed to inform targeted policy interventions. (3) Climate change impacts on RWH reliability. Projected changes in precipitation intensity, frequency, and seasonality under climate change will significantly affect the performance and reliability of RWH systems. High-resolution climate projections coupled with hydrological models are required to assess future RWH potential under alternative climate scenarios (SSP1-2.6 through SSP5-8.5). This research is critical for adaptive design and planning. (4) Integration of RWH with precision agriculture. The convergence of low-cost sensors, real-time weather forecasting, and smart irrigation controls presents unprecedented opportunities to optimize RWH system

performance. However, research on the integration of sensor-based irrigation scheduling with harvested rainwater systems—particularly in smallholder contexts—is still in its infancy. Field-scale experiments evaluating the synergistic effects of RWH, sensor-based irrigation, and real-time decision support systems are urgently needed.

Author contributions

YW: Software, Investigation, Writing – original draft, Methodology. JP: Writing – original draft, Formal analysis, Conceptualization. ZL: Data curation, Writing – original draft. RW: Writing – original draft. XW: Writing – original draft, Resources, Software, Writing – review & editing, Funding acquisition, Conceptualization.

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Supplementary material

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